

Digital Material Passport

ID TEST-SPECIMEN-001 - Version 0.1.0
Issue Date 2025-09-29 - Certificate Type

Manufacturer	Customer	Business Transaction	
Steel Testing Labs GmbH 123 Test Street 10115 Berlin DE	Customer Industries Inc 456 Customer Ave New York 12345 US	Order	ORD-2025-001
		Delivery	DEL-2025-001 · 1000 kg

Product

Product Name	High Strength Steel Plate	Shape	Plate - Width 2000 mm - Thickness 50 mm - Length 6000 mm
Batch ID	BATCH-2025-001		

Chemical Analysis

Heat Number H123456 - Melting Process EAF+LF - Casting Method ContinuousCasting - Casting Date 2025-09-28

Symbol	Unit	Min	Max	Actual	Symbol	Unit	Min	Max	Actual
C	%		0.2	0.18	Ni	%		0.2	0.15
Mn	%	1.4	1.7	1.45	Mo	%		0.1	0.08
Si	%		0.5	0.35	Cu	%		0.25	0.18
P	%		0.025	0.012	Al	%	0.02	0.05	0.025
S	%		0.015	0.008	N	%		0.012	0.008
Cr	%		0.3	0.25	CEV	%		0.45	0.42

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.42%

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength ¹ Specimen: 1/4T, L	Rm	485 MPa	450 MPa	600 MPa	ASTM E8	✓
Charpy V-Notch Impact Temperature: -40°C Specimen: 1/4T, L-T	KV	48 / 52 / 45 J - Average 48.3 - Min 45 - Max 52 - Std Dev 3.5	40 J		ASTM E23	✓
Through-Thickness Tensile Room temperature, through-thickness direction					EN 10164	✓

Location (position)	0	0.25	0.5	0.75	1
Value [MPa]	475	485	490	487	478

Yield Strength ¹ Specimen: Surface, T - ID: YS-SURF-T-001	Re	355 MPa	335 MPa	ASTM E8	✓
Elongation ¹ Specimen: Custom (Mid-radius at end section), C - ID: ELONG-MR-001	A	22 %	20 %	ASTM E8	✓

¹ Room temperature

Validation

We certify that the material described herein has been tested and inspected in accordance with the specified standards and meets all requirements.

Validated by Dr. Hans Schmidt, Quality Manager - 2025-09-29